

AVISHEK SARKAR

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Experience & Leadership

Research Assistant – Department of CSE, JKKNIU

Feb 2025 – Present

- Working in a 4-member research team on adaptive fingerprint recognition systems
- Developing an anti-aging fingerprint recognition system with automatic template updating over time
- Studying fingerprint aging from skin changes, sensor variation, and bio-metric drift

Senior Executive Committee Member – Mid-Day Programming Club, JKKNIU

Sep 2024 – Oct 2025

- Led and mentored a 5-member team to organize 20+ contests, tech fests, and workshops
- Managed logistics and promotion for events with 50–300+ participants
- Boosted club engagement by 30% and helped grow membership from 60+ to 100+

Education

Bachelor of Science (B.Sc.) in Computer Science and Engineering

Jatiya kabi kazi Nazrul Islam University

Feb 2022 – Jan 2026

- CGPA: 3.67 / 4.00, Ranked 1st in batch

Thesis & Projects

Sleep Stage Classification Using Machine Learning (*Undergraduate Thesis*)

- Processed ECG signals from SHHS visit-2 dataset for multi-class sleep stage classification
- Applied machine learning and signal processing techniques without using any augmentation
- Handled class imbalance with feature engineering, class weights, and ensembles to improve accuracy

Job Recommendation System (*GitHub Repo*)

- Built a full-stack AI-powered job recommender with TF-IDF, cosine similarity, and weighted scoring
- Extracted skills, degrees, experience, and locations via hybrid NLP parsing (spaCy, regex, PyMuPDF)
- Engineered real-time recommendations (<2s) with a modular Flask architecture and scikit-learn

Technical Skills

- **Languages:** Python, C/C++, Java, PHP, JavaScript, SQL
- **AI / ML & Data Science:** scikit-learn, spaCy, NumPy, pandas, TensorFlow
- **Web Development:** Flask, Jinja2, HTML/CSS, Bootstrap, RESTful APIs, Socket.IO
- **Tools & Databases:** MySQL, SQLite, Git, GitHub, PyMuPDF, CLI, Tableau
- **Hardware & Others:** Arduino, IR & Ultrasonic Sensors

Achievements

- **Annual Merit Scholarship Recipient at JKKNIU (2022–2025)** – Awarded consecutively for academic excellence
- **4th Place in TechMind Hackathon 2025 at JKKNIU** – Ranked 4th out of 25 teams; developed *MealMate*
- **5th Place in Line Follower Robot Competition at UIU CSE Fest 2025** – Ranked in top 9% out of 55 teams; among 18 finalists
- **14th Place in Capture the Flag Competition at BUP CSE Tech Carnival 2025** – Ranked 14th on-site out of 40 teams; 11th in preliminary round out of 120 teams
- **Onsite Finalist at NASA Space Apps Challenge 2024** – Selected in top 10% from 500+ teams; developed *GlobeGuardian*